

BBJ Sense Industrial SCADA & IoT Platform

Complete Software Design Summary

Current Version: v0.3.1

Project Owner: BBJ Electricals

PROJECT VISION

BBJ Sense is a cloud based Industrial IoT and SCADA platform designed for industrial plants, electrical panels, solar plants and energy monitoring systems.

Core capabilities:

- Real time monitoring
- Industrial device connectivity
- Energy management
- Alarm management
- Reports
- Multi tenant customer management

TECHNOLOGY STACK

Backend:

- Python 3.12
- FastAPI
- SQLAlchemy ORM
- Alembic migrations
- PostgreSQL 16
- JWT Authentication
- Pydantic
- Uvicorn

Deployment:

- Google Cloud Platform
- Compute Engine VM
- Systemd service: bbjsense

SYSTEM ARCHITECTURE

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FastAPI Backend

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PostgreSQL Database

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Tenant Management

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Users / Sites / Gateways / Devices / Telemetry / Alarms

MULTI TENANT ARCHITECTURE

Implemented.

Each customer is isolated using tenant_id.

Current tenant:

BBJ Electricals

Tenant ID: 1

Isolation implemented for:

- Users
- Devices
- Telemetry
- Gateways

SECURITY SYSTEM

Implemented:

- JWT login
- Role based access control
- Permission system
- Gateway API key security
- Tenant data protection

USER ROLES

Supported:

- Super Admin
- Engineer
- Client Admin
- Plant Manager
- Operator
- Viewer
- Maintenance Engineer

DEVICE MANAGEMENT

Implemented:

- Device registration
- Device configuration
- Modbus RTU support
- Device tags
- Dynamic templates

Current tested device:

ABB M1M12 Energy Meter

Slave ID: 2

COM6

4800 Baud

Odd Parity

2 Stop Bits

TELEMETRY SYSTEM

Implemented monitoring:

- Frequency
- Voltage
- Current
- Active Power
- Energy
- Power Factor

Gateway flow:

Industrial Meter

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RS485 Modbus

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Gateway Agent

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Secure API

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BBJ Sense Cloud

GATEWAY MANAGEMENT

Implemented:

- Gateway credentials
- Secure gateway keys

- Tenant linked gateways
- Gateway telemetry API

ALARM SYSTEM

Implemented foundation:

- Alarm scheduler
- Frequency alarms
- Voltage alarms

Future:

- Over current
- Breaker trip
- Device offline
- Communication failure

GIT VERSION HISTORY

v0.3.1 - Gateway tenant isolation

v0.3.0 - Telemetry tenant isolation

v0.2.2 - Device tenant migration and JWT tenant support

v0.2.1 - User tenant isolation

v0.2.0 - Multi tenant foundation

ROADMAP

v0.3.2

Gateway health monitoring

v0.4

Next.js SCADA dashboard

v0.5

Industrial Windows Gateway Agent

v0.6

Flutter mobile application

CURRENT STATUS

Completed:

- Backend foundation
- Database architecture

- Security
- Multi tenancy
- Device management
- Telemetry pipeline
- Gateway architecture

Pending:

- Frontend SCADA dashboard
- Mobile application
- Industrial gateway software

CONCLUSION

BBJ Sense has achieved a strong industrial IoT SCADA foundation with enterprise architecture including tenant isolation, secure authentication, device management, gateway communication and telemetry processing.